

Signal Analysis & Telemetry Extraction Report

Target Recording: TEST_FSK_TELEMETRY_BEACON.wav | Format: WAV | Sample Rate: 48.0 kSps

SIGNAL HEALTH
74/100
STATUS: WARNING

ANALYST PRIORITY
MEDIUM
Warning state or
moderate...

CONFIDENCE
94.4%
DSP DATA QUALITY

ANOMALIES
1
DETECTED ISSUES

1. Explainable Analysis & Executive Summary

Automated analysis completed for 'TEST_FSK_TELEMETRY_BEACON.wav'.

The recording contains 6 detected active segment(s) across a total duration of 2.000s. Spectral analysis identifies a center frequency at 5.375 kHz with an occupied bandwidth of 12.352 kHz and an SNR of 39.4 dB (Modulation profile: Complex Telemetry / Multi-carrier).

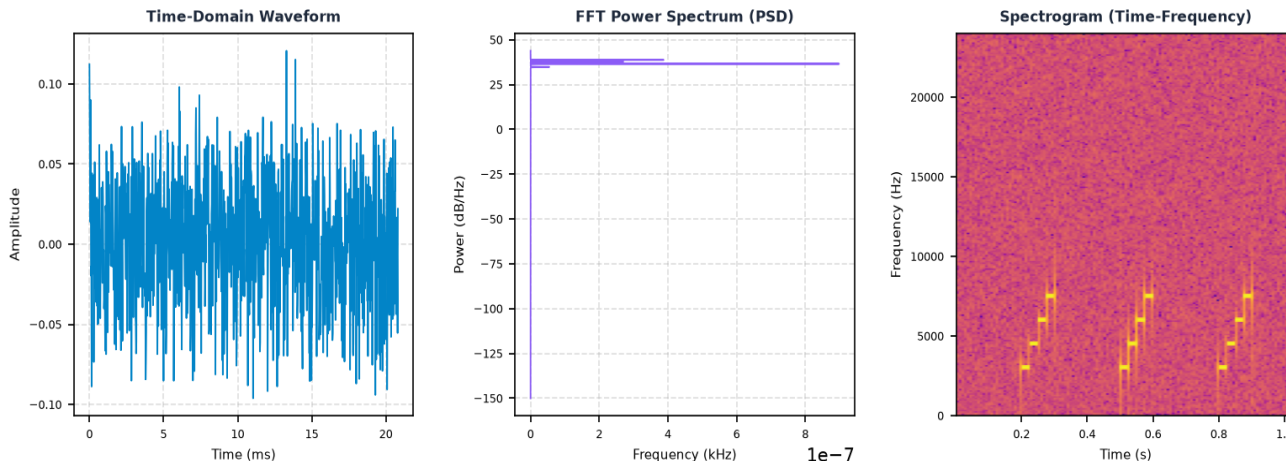
Benchmarked against historical baseline 'S-BAND TELEMENTARY '. Statistical drift analysis indicates baseline center frequency was 5375.3 Hz (observed: 5375.3 Hz) and nominal SNR was 39.4 dB.

Detected 1 anomaly condition(s):

- Segment #3 Profile (WARNING): Segment #3 at 0.796s deviates significantly in power/duration envelope compared to adjacent pulse segments.

Overall Signal Health Score is 74.0/100 (WARNING), assigning this file to MEDIUM PRIORITY for engineer triage.

2. Time-Domain, Spectral & Time-Frequency Visualization



3. Extracted Signal Parameters

MEASURED PARAMETER	VALUE	DERIVED / ESTIMATED PARAMETER	VALUE
Peak Frequency	3.000 kHz	Occupied Bandwidth (99% OBW)	12.352 kHz
Spectral Centroid	5.375 kHz	-3 dB Bandwidth	46.88 Hz
Signal Power (dBFS)	-9.58 dB	-10 dB Bandwidth	93.75 Hz
Est. Noise Floor	-72.07 dB	Signal Stability Index	0.303 / 1.0
Measured SNR	39.4 dB	Crest Factor / PAPR	3.012 (9.58 dB)

Peak Amplitude	1.0 V	Spectral Flatness (Entropy)	0.0294
RMS Amplitude	0.332 V	Modulation Classifier Hint	Complex Telemetry / Multi-carrier (75.0%)

4. Signal Fingerprint & 'What Changed?' Analysis (Fingerprint ID: FP-CF5BB6DE970B9B48)

Parameter	Baseline Mean	Current Value	Delta (%)	Dir	Status
Center Frequency	5375.27 Hz	5375.27 Hz	+0.0%	→	Nominal
Bandwidth (99% OBW)	12351.56 Hz	12351.56 Hz	+0.0%	→	Nominal
Signal-to-Noise Ratio (SNR)	39.4 dB	39.4 dB	+0.0%	→	Nominal
Signal Power	-9.58 dBFS	-9.58 dBFS	+0.0%	→	Nominal
Signal Stability	0.303 index	0.303 index	+0.0%	→	Nominal
RMS Amplitude	0.332 V	0.332 V	+0.0%	→	Nominal

5. Anomaly Investigation Log (1 events)

Severity	Affected Parameter	Observed Value	Engineering Reason
WARNING	Segment #3 Profile	Power: -4.77 dB, Dur: 0.109s	Segment #3 at 0.796s deviates significantly in power/duration envelope compared to adjacent pulse segments.